ERDPlus – instructions

Databases

Academic year 2020./2021.

Programski alat ERDPlus

The programming tool ERDPlus acts as a help in modelling a relational database.
ERDPlus is a free online tool and doesn't have to be installed locally on the computer.

Main features of the ERDPlus tool which will be mostly used on the *Databases* course are:

- creation of the ER-diagrams and relational diagrams (relational shemas)
- automatically mapping the ER-diagram to the relational schema
- export of the standard SQL DDL queries based on the relational schema
- export of the diagrams in the PNG format.

This tool will be used in the 3rd homework and during the 2nd lab work.

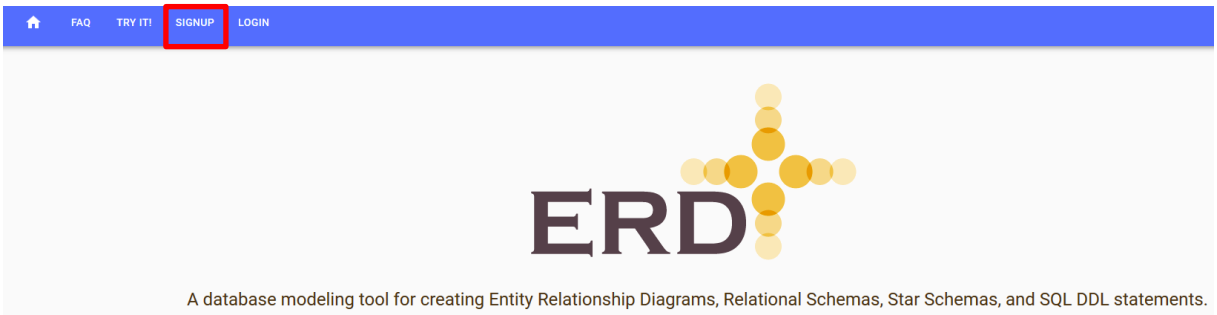
Creating a user account

The tool can be used with the basic (limited) fuctionalities as an anonymous used, although in that case the diagrams can be saved and later used. In case a registered user with a user account uses the tool gives the option of saving the created diagrams to the server. By creating a user account and without logging in, the tool can be used with the limited functionalities.

The web page of ERDplus is https://erdplus.com/.

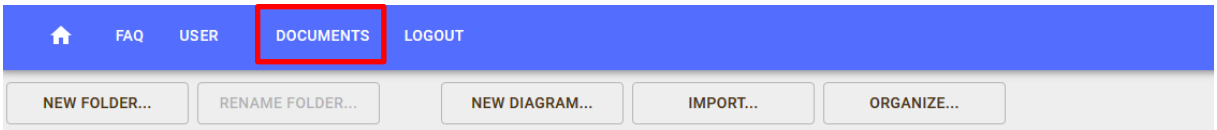
As a broader functionalities are needes for this course, it is necessary to create a user account.

To create a user account, click on the *SIGNUP* option in the menu.

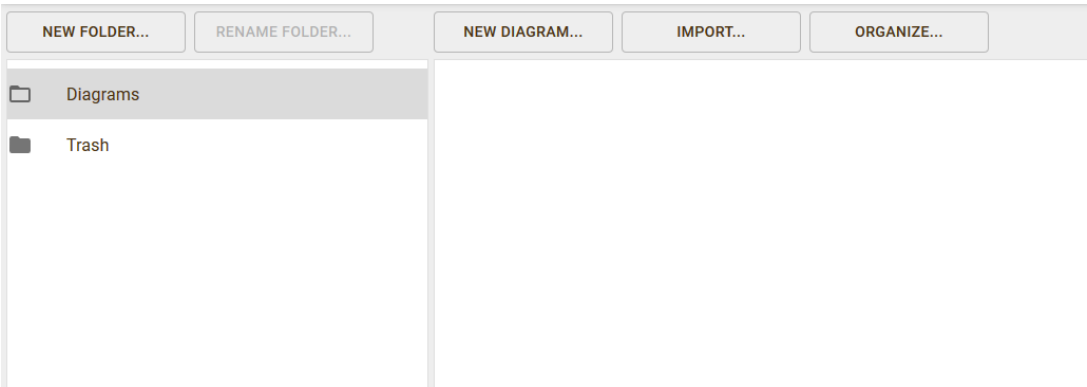


After which the user data have to be entered and finalized by clicking on the *CREATE ACCOUNT*.

Upon the user account creation, the user is logged in the system and redirected to its own storage space (*DOCUMENTS*) for saving the documents on the server.



On the left the folders are shown, while on the right the content of the chosen folder are visible.



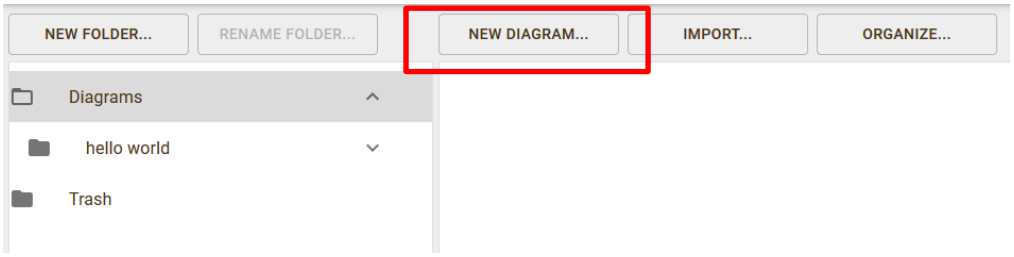
Inside its own storage, the user can create, rename and edit diagrams and folders.

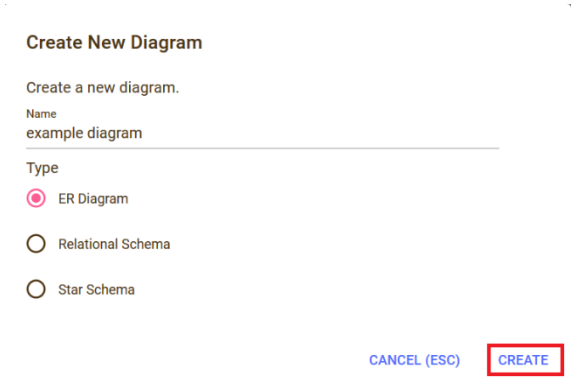
The created diagrams can be imported and exported as files. By clicking on the button IMPORT, the locally stored digrams can be imported to the chosen folder in the tool.

Working with diagrams and files

Creating the diagram

The diagram is created by clicking on the button *NEW DIAGRAM…* after which the diagram name should be entered. As the ER diagram has to be created in the example, choose the option *ER Diagram* and click *CREATE.*

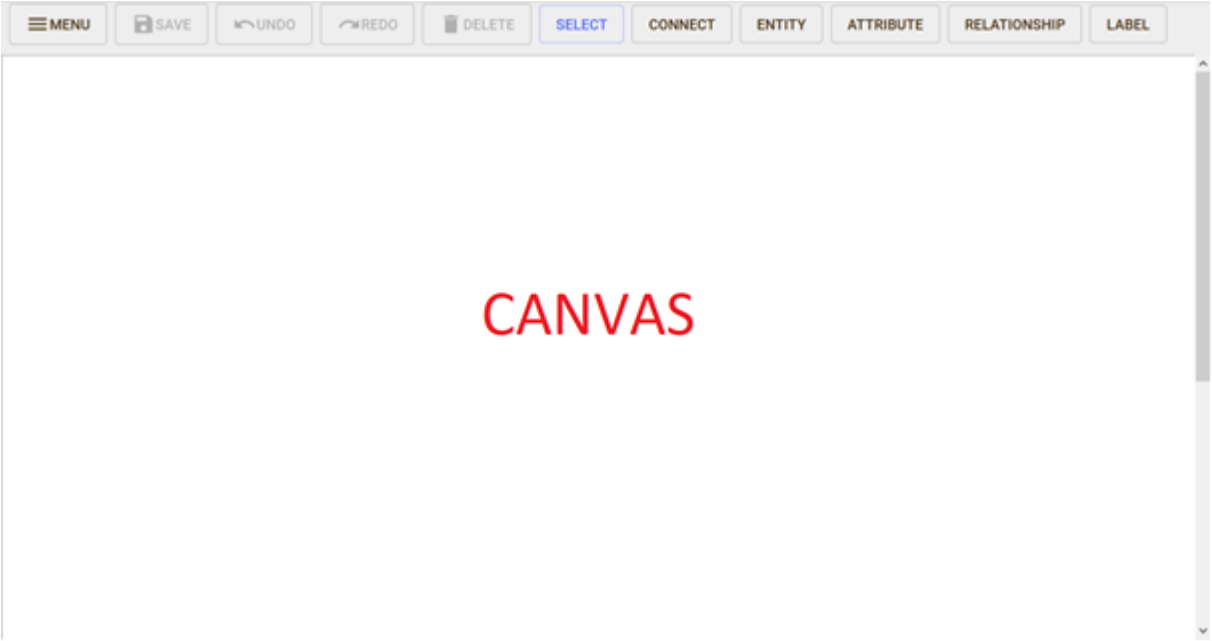




The effect of this action in the user space is similar to the action „New File“ in the filesystem; a new ER –diagram is created in the chosen folder („Diagrams“).

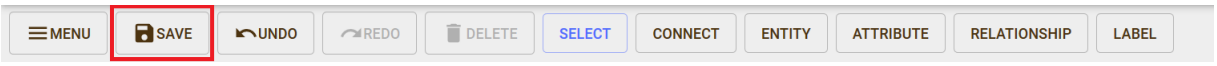
For the newly created diagram, by choosing the three dots on the far right (⋮) an activity menu for the chosen diagram is shown

By choosing the option *Open…* the editor for the chosen diagram is opened that currently contains the empty drawing canvas and the toolbar.



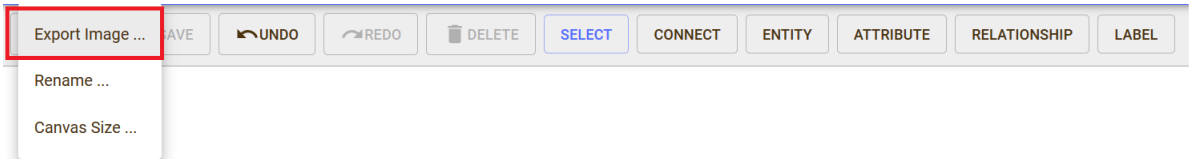
Saving the changes

Although the tool often saves the changes made in the active diagram automatically, the changes can be stored manually by clicking on the button SAVE. In case the button is disabled, either there were no changes since the last diagram import or the tool already saved the changed automatically.



Exporting the diagram as an image

On the toolbar choose the button *MENU* > *Export Image…* > *OK*, after which the exported diagram image in the *.png* format will be available for the download.

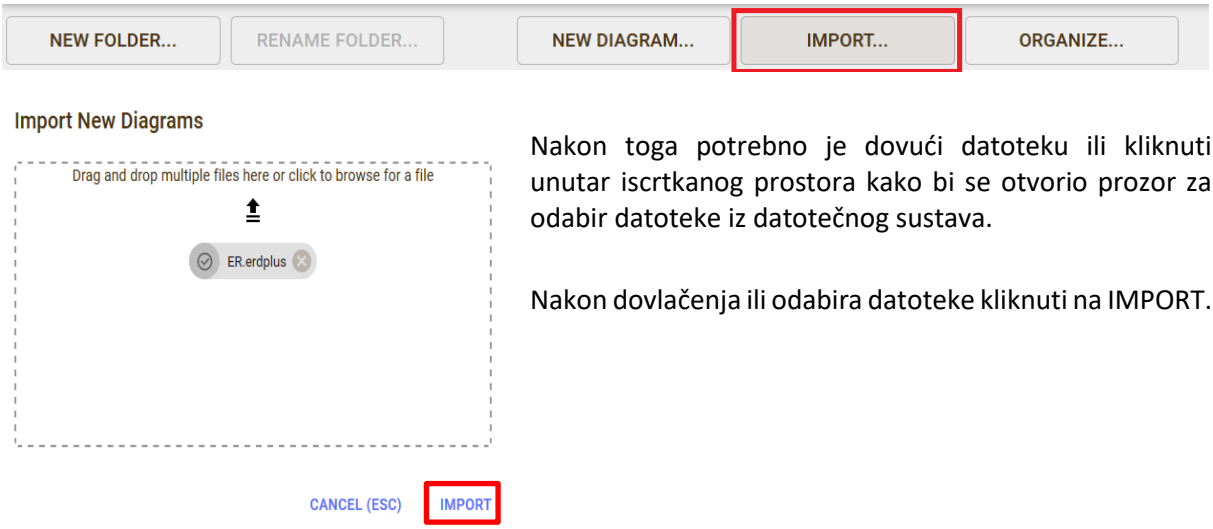


Exporting the diagram as ERDPlus file type

To export the diagram in the ERDplus file type, the user should go to the user space (by clicking on the button *DOCUMENTS*). For the chosen diagram the activity menu has to be activated (clickin on the ⋮ on the right), followed by the options *Export* > *OK*, by which the file of the file type *.erdplus* will be available for the download.

Importing the diagram from the existing ERDPlus file

To import the diagram as ERDPlus datoteke the user has to be in the user space (DOCUMENTS) and choose IMPORT….



Nakon toga potrebno je dovući datoteku ili kliknuti unutar iscrtkanog prostora kako bi se otvorio prozor za odabir datoteke iz datotečnog sustava.

Nakon dovlačenja ili odabira datoteke kliknuti na IMPORT.

Examples – part 1

The examples will guide you through the steps of creating the diagram for the simple database, where each step assumes that the prior steps were mastered. Due to that it is advisable to go through the instructions sequentially, without skipping any of the steps.

Task 1.

Create a simple ER-dijagram for the database which records the data about the students and their activities. Each student record includes JMBAG (student identification number), name and surname, and for activity its code and name. The database records whether a student takes part in an activity;

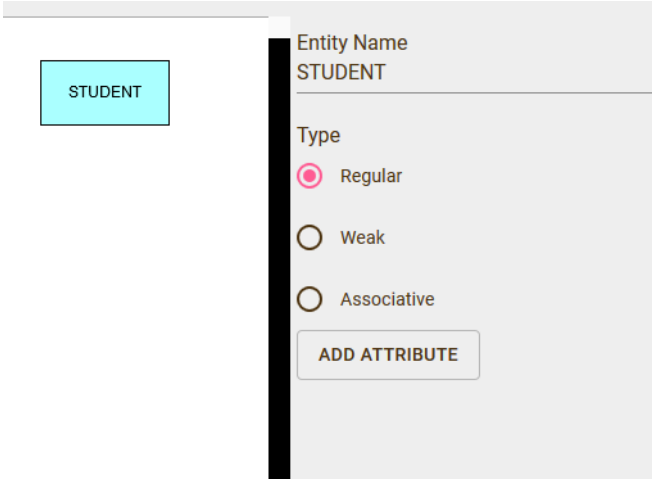
student can take part in zero or more activities, while zero or more students can take part in a single activity.

Entities

To add an entity the button *ENTITY* from the toolbar should be chosen, after which the user should click anywhere on the canvas. An editor for the new entity is opened automatically on the right side, where the entity name should be entered (e. g. STUDENT) and its type chosen. In this case, the chosen entity is regular (not weak).

Generally, from the suggested options you will need:

- **Regular**: Regular entity
- **Weak**: Weak entity



The same steps should be repeated for the entity ACTIVITY.

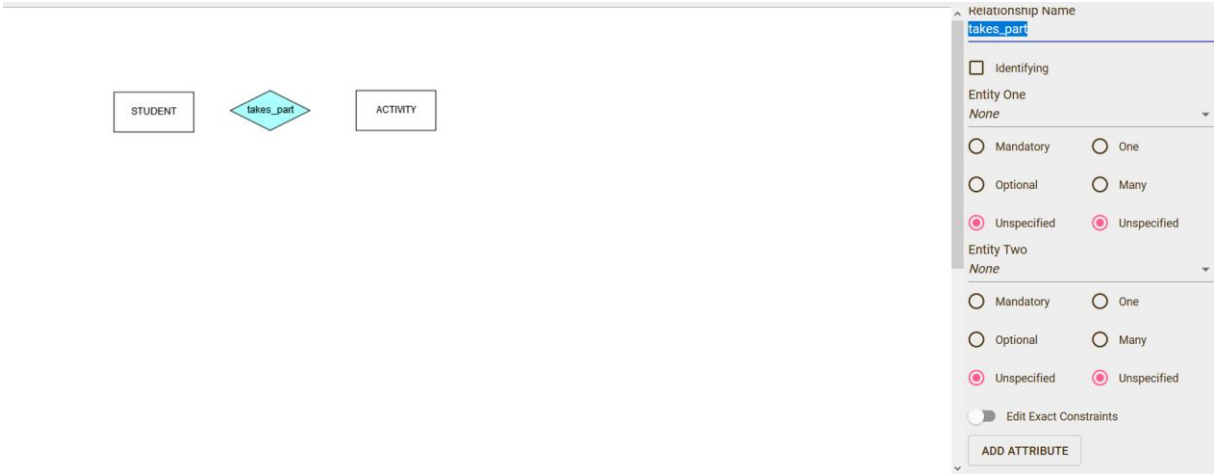
Note:

- Entites can be arranged in the canvas through *drag-and-drop*. Moreover, some or all of the elements in the canvas can be selected (chosen), and the chosen elements can be arranged by dragging over the canvas. To enable the selection, the option ***SELECT*** on the toolbar should be chosed, which will set the tool in the SELECT mode. More elements on the diagram can be chosen (selected) in two ways:
  - Drag the rectangle using the mouse which will „capture“ all the selected elements
  - While holding the CTRL key on the keyboard, clicking the mouse on one element at the time.
- If only one diagram element inside the canvas is chosen (selected), the tool will mark it blue and will open its editor automatically

- For the chosen diagram element:
  - The key *Delete* on the keyboard deletes its name
  - Choosing *DELETE* on the toolbar deletes the selected entity.

Binary N:N relationship

To add a new relationship, the button *RELATIONSHIP* from the toolbar should be selected, followed by clicking anywhere inside the canvas. On the right the new relationship editor is automatically opened.

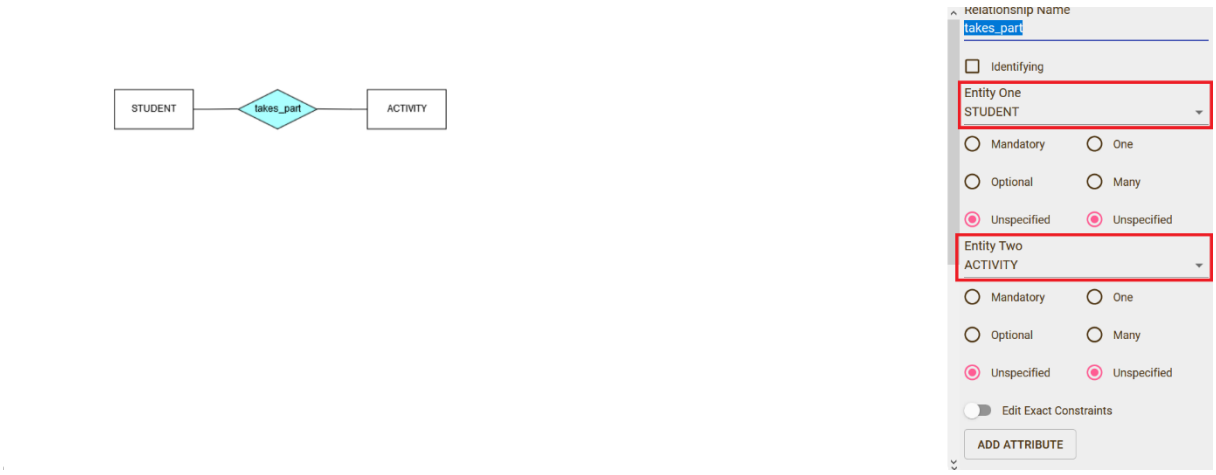


Followed by entering the name of the relationship (e. g.. „takes_part“).

The next step is connecting the relationship to the entities.
It can be done in two ways:

<u>Method 1:</u> Using the dropdown menus.

Activate the relationship editor. From the dropdown menus manually set the entities *Entity One* i *Entity Two*, after which the connecting lines between the relationship and the chosen entities will be shown.



<u>Method 2:</u> Using the *CONNECT* option.

Choose the *CONNECT* on the toolbar. Click either on the entity STUDENT or on the relationship and while **holding** the left mouse button– drag the line between them. When the relationship, entity and the connecting line go green, release the mouse button.

Note:

After creating the first connection betwen relationship and the entity, the tool isn't in the *CONNECT* mode anymore.
Before creating the second connection the *CONNECT* option should be selected again.

Repeat the procedure for the entity ACTIVITY.

Upon connecting of the relationship with the entities STUDENT and ACTIVITY, in the relationship under *Entity One* the label STUDENT is shown, while in the *Entity Two* the label ACTIVITY is shown. In that way the tool automatically sets the entities which the relationship is connected to.

Any of the two mentioned methods may be used to connect the relationship and entities.

The next step is entering the entity cardinality.

Choose the relationship to activate the editor on the right side.

The relationship in the focus is the binary N:N relationship (student can take part in several activities, and multiple students can take part in a single activity). Therefore, the upper boundary for the both relationship ends will be „many“. But what about the lower boundary? Does a student necessary take part in a activity? Should all activities have to be taken by at least one student? In both cases the answer is „no“, as it „zero or more“ constraints have been specified.

Therefore, set the entity cardinality settings – the upper cardinality setting will be *many*, while the lower cardinality setting is zero for <u>both</u> entities connected by the relationship.